Home lab week 11, Day 4

Correct identified firewall or segmentation issues and repeat the relevant tests to verify that the security controls are working as intended.

The image shows a Wireshark packet capture detailing network activity from a host at `192.168.10.10`.

The logs capture active network reconnaissance, including host discovery (ping scanning) and targeted communication attempts that are failing to receive responses.

🔍 Detailed Traffic Breakdown

1. Targeted Communication & Probing (Packets 1–8)

- **The Source:** A single host at `192.168.10.10`.
- **The Target:** A specific destination at `192.168.20.10` (located on a different subnet).
- **The Activity:**
  - Sent ICMP Echo (ping) requests and ICMP Timestamp requests.
  - Attempted TCP connections to port 443 (HTTPS) via a `[SYN]` packet and port 80 (HTTP) via an `[ACK]` packet.
- **The Outcome:** Every single one of these packets is flagged with **`(no response found!)`**. The target is either offline, blocking the traffic with a firewall, or there is a routing issue preventing the return path.

2. Local Gateway Resolution (Packets 9–10)

- **The Activity:** An Address Resolution Protocol (ARP) exchange occurs on the local network.
- **The Details:** `192.168.10.10` broadcasts a request asking *"Who has 192.168.10.1?"* (likely its default gateway). The gateway responds, mapping its IP to the MAC address ending in `d1:26:3d`.
- **The Significance:** This confirms that local layer-2 and layer-3 communication on the source's immediate subnet is functioning perfectly.

3. Subnet Ping Sweep / Network Scan (Packets 11–17+)

- **The Activity:** A rapid succession of ICMP Echo requests (pings) sent in a span of mere milliseconds (around the 12.52-second mark).
- **The Pattern:** The source systematically pings sequential IP addresses: `192.168.1.1`, `192.168.1.2`, `192.168.1.3`, `192.168.1.4`, and so on.
- **The Outcome:** Except for packet 11 (which notes a reply later in packet 21), these targets show `(no response found!)`.
- **The Significance:** This behavior is characteristic of an automated **network ping sweep** (host discovery scan) typically performed by tools like Nmap, Angry IP Scanner, or malicious actors mapping out an environment.
